Integration of Phenolic Secondary Metabolism into a Grapevine Genome-Scale Metabolic Model Using DeePlants

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Abstract. Phenolic compounds are the main group of secondary metabolites and play an important role, both in the defence and environmental adaptation of grapevine (*Vitis vinifera*) and in wine quality. However, their biosynthesis is frequently underrepresented in metabolic databases and genome-scale metabolic models (GSMMs). This work studies the phenolic secondary metabolism of the grapevine GSMM iMS7199 using the DeePlants tool, a deep-learning framework for the elucidation of secondary metabolism. Fourteen phenolic compounds present in a dataset but absent from the iMS7199 GSMM were selected and subsequently characterised through DeePlants' pathway prediction (KEGG, PlantCyc and NP-Classifier) and the prediction of retrobiosynthetic reactions. The grapevine proteome (PN40024.v4) was functionally annotated with the ProtBERT protein language model to predict EC numbers, and candidate enzymes were assigned to each predicted reaction.

The case analyses of benzoic acid and viniferin illustrate the strengths and limitations of the approach, since the structural classification was generally accurate, whereas the *de novo* predicted pathways ranged from biologically inconsistent, proposing precursors and enzymes without precedent in plant metabolism, to metabolically plausible at the metabolite level but enzymatically oversimplified, frequently omitting multi-step transformations. Likewise, the pathway classification tended to associate compounds with adjacent or analogous pathways rather than the dedicated one. These results highlight both the potential and the current limitations of automated retrobiosynthesis for the expansion of underrepresented secondary metabolic pathways in plant GSMMs.

Keywords: Phenolic compounds · Retrobiosynthesis · Deep Learning.

1 Motivation

In grapevines, secondary metabolites are generally not considered essential for basic plant growth, but are crucial for its defence and, from an agricultural point of view, important for wine quality [7]. These compounds are primarily affected by the environment, such as light and temperature and by viticultural practices [11]. Genome-scale metabolic models (GSMM) are important tools used to represent plant's metabolic network. The study will involve the *Vitis vinifera* GSMM iMS7199, featuring secondary metabolites such as anthocyanins, quercetin, resveratrol, terpenoids, carotenoids, cholesterol, and diacylsucrose [17]. Through the integration of the model, it becomes possible to examine how the plant allocates its metabolic resources towards the biosynthesis of phenolic compounds. To predict how the plant redistributes its resources to produce phenolic compounds, the main group of plant secondary metabolites that highly contribute to wine flavor, the model will be integrated with a computational framework, DeePlants, aiming to improve the elucidation of secondary metabolism, as well as the identification and functional mapping of secondary metabolites and biosynthetic genes. This tool will allow the expansion of metabolic pathways that are poorly represented in databases. To this end, the aim of the present work is to use DeePlants to study the phenolic compounds of the grapevine that are absent from its Genome-Scale Metabolic Model, predicting their biosynthetic routes through retrobiosynthesis and assigning EC numbers and candidate enzymes to the predicted reactions.

2 State of the art

2.1 Plants secondary metabolites

Plants yield a vast array of secondary metabolites that play a fundamental role in mediating environmental interactions and stress adaptation [21], with over 200,000 different known compounds [20]. Classified into three main groups, terpenoids, alkaloids, and phenolic compounds (PCs) [3], they play crucial roles in plants, such as regulation of growth and development, and respond to diverse biotic and abiotic stresses, such as pathogens and herbivores [21]. Unlike primary metabolites, their production is typically triggered by specific environmental stimuli or during specific developmental stages, increases their chances of survival [21]. The current literature demonstrates that plants use these compounds to manipulate their surrounding environment through plant-plant, plant-animal, and plant-microorganism interactions [20]. Despite exhibiting diverse biochemical properties, the best characterized and most widely highlighted is their antioxidant activity [5]. If antioxidant enzymes cannot sufficiently neutralize oxidation, secondary metabolites such as PCs become critical for limiting reactive oxygen species (ROS) accumulation [21].

2.2 Phenolic compounds

PCs are generally present in most plant tissues, such as fruits, seeds, leaves, stems, roots, among others[16]. They are composed of at least one aromatic ring with a hydroxyl group [16] and present great structural diversity, exceeding 8,000 PCs overall [16]. They can be divided, according to their structure, into the groups of phenolic acids, flavonoids, tannins, phenolic lignans and phenolic stilbenes [24]. PCs represent one of the largest groups of secondary metabolites in the plant kingdom [10], playing fundamental roles in plants, such as providing structural support and protection against UV radiation, biotic and abiotic stress. They also play a crucial role from a consumer perspective, through their antioxidant activity, as well as through their involvement in the regulation of various cellular processes [16]. Furthermore, they influence the quality attributes of fruits and vegetables, including bitterness, colour, and flavour [16].

In plants, PCs biosynthesis proceeds via shikimic acid pathway [23], where carbohydrates from glycolysis and the pentose phosphate pathway are converted into chorismate, a precursor of tryptophan and phenylalanine. In the case of phenylalanine, it initiates the phenylpropanoid pathway through a deamination reaction catalyzed by phenylalanine ammonia-lyase (PAL), yielding *trans*-cinnamic acid. This enzymatic reaction represents a crucial regulatory step in the production of numerous phenolic compounds. Intermediate metabolites in this pathway are also converted into phenolic acids [13].

Over the last decade, at least 138 PCs have been identified in the vegetative organs of the *Vitis vinifera*, including stilbenes, anthocyanins, hydroxycinnamic acids, hydroxybenzoic acids, flavanones, flavones, flavonols, flavan-3-ols, and coumarins. These compounds are distributed throughout the grapevine, with flavonols and flavan-3-ols being the most represented in the leaves and stems [8]. In grape berries, PCs are primarily present in the skins and seeds, and flavonols are the most abundant PCs in grape skins [12].

PCs contribute to the perception of wine quality, influencing the color, for browning reactions as well as for various reactions with anthocyanins that lead to color stabilization in red wines [12][22], flavor, influencing sensory attributes like bitterness and astringency [4], and sensory characteristics of the wine. Polyphenols, particularly certain phenolic acids and flavonols, participate in the copigmentation phenomenon, which is why anthocyanins exhibit a much more intense color than would be possible without them [12][22].

2.3 *Vitis vinifera* model

Through the integration of tissue-specific omics data for the grapevine leaf, stem, and grapes, a diel multi-tissue model of *V. vinifera* was constructed by Sampaio and collaborators [17].

The study involves the genome-scale metabolic model (GSMM) iMS7199, specifically reconstructed for the grapevine *V. vinifera*. The genome version PN40024.v4 was used for this study. This model presents 5399 reactions with 1624 transporters and 244 exchanges, and 5141 metabolites, distributed across eight compartments, these being cytosol, mitochondria, Golgi apparatus, endoplasmic reticulum, chloroplast, peroxisome, vacuole, and extracellular space[17]. Since genomic annotation was performed using protein sequences, the Gene-Protein-Reaction (GPR) rules in this model were established using protein identifiers rather than gene identifiers. The model includes 7199 protein identifiers that represent the 6018 genes of *V. vinifera* genome [17].

The secondary metabolite biosynthesis, derived from the repository containing metabolic information from the PlantCyc 14.0 and MetaCyc 26.1 databases, and the Universal Protein Resource (UniProt), is the metabolic pathway class associated with the highest number of unique reactions, around 140. Compared to previous plant models *V. vinifera*'s GSMM represents a major advancement, especially for secondary metabolism, which is frequently underrepresented in plant models [17].

Through this model and using the integration of RNA-Seq data, specific models were created for different parts of the plant, such as the stem, the leaf, and the grape berry, with the latter being further divided into green and mature developmental stages. To make the simulation more realistic, the models were merged into a diel multi-tissue model, which simulates the light and dark phases. The model allowed the simulation and investigation of the grapevine's metabolic behavior under different phototrophic and heterotrophic conditions. With the aim of identifying the associated metabolic pathways and gene annotation, these reactions were analyzed and secondary metabolite biosynthesis pathways were the ones that obtained the highest number of unique reactions. The main limitation is the lack of experimental data to define pseudo-reactions representing the consumption of macromolecules for biomass synthesis. The model can be integrated with other datasets to investigate condition-specific metabolic phenotypes, providing a better understanding of plant metabolism [17].

FBA and pFBA Methods such as Flux Balance Analysis (FBA) and parsimonious FBA (pFBA) are required to verify the ability of the GSMM iMS7199 to accurately simulate key metabolic processes, including photosynthesis, photorespiration, and respiration[17].

FBA is a metabolic modelling approach that uses a set of physical, biochemical, and thermodynamic constraints to predict steady-state metabolic fluxes from stoichiometric models [18], where stoichiometric models are the mathematical representations of the organism's metabolic reactions, in which the model assumes a steady state, where the rate of change of metabolite concentrations over time is zero, and metabolic fluxes describe the rates at which each metabolite is consumed or produced by each reaction in the network[14]. Since it is based on linear equations, FBA is computationally efficient and widely used to simulate flux distributions in extensive metabolic networks. This is commonly used in the study of plant metabolism [18], and through these calculations, it is possible to determine an optimal flux distribution for a given objective function [17]. This has been widely used to simulate GSMMs. However, in the solution space for a particular objective, there are generally multiple optimal solutions. In this model, pFBA is employed to simulate plant secondary metabolism, an extension of FBA which, by selecting a flux distribution from the optimal FBA solution space, minimizes the total sum of fluxes [17].

2.4 DeePlants

Retrobiosynthesis The biosynthetic pathways of more than 90% of natural products (NPs) are still undiscovered. Despite their complex and highly diverse structures, NPs are synthesised from a few simple biological building blocks. For this purpose, retrobiosynthesis is employed, which starts from a specific target molecule and aims to identify a viable and efficient synthetic pathway, going through a large number of possible chemical reactions until ultimately reaching the precursor molecules [25]. Given the performance of reinforcement learning, the use of Monte Carlo Tree Search (MCTS) guided by neural networks was proposed for retrosynthetic planning. This approach is fundamental, as it enables the creation of a self-improving learning process for the neural networks guiding the search algorithms, allowing them to assess a given state and predict the most efficient and viable reactions to attempt next [19][25].

Identifying these synthetic routes is crucial and computer-assisted retrosynthesis is essential to solve it. The MEEA* algorithm offers a powerful solution by combining the heuristic algorithms MCTS and A*, allowing it to escape local optimal branches and preventing exhaustive exploration of all branches [25].

MCTS operates through a balance between exploitation, where it takes advantage of and expands upon paths that have already been tested and which, according to current scores, appear to be the most promising and efficient, and exploration, where new and unknown paths are tested in the research tree, ensuring that potentially better alternative routes are not overlooked. The algorithm operates through a search process that iterates through four steps. The first is selection, where the algorithm traverses the search tree, used by algorithms to map the various possible chemical synthesis pathways of a molecule, through two parts, the nodes and the edges. The nodes represent the states of the synthesis. The initial node, the root of the tree is the target molecule and the nodes below it contain the set of intermediate molecules and building blocks present at that point in the synthesis. The edges are the connections between the nodes. Each edge represents a chemical reaction that causes the transition from the parent node, the products, to the child node, the reactants. This step uses an exploratory tree policy like pUCT, which calculates and ensures a balance between exploring new options and exploiting those that already appear promising. The second step is the expansion, where new promising nodes, suggested by a retrosynthetic model that predict the most promising chemical reactions for breaking down a target molecule into its possible precursors, are incorporated into the tree. The third step is evaluation, estimating the subsequent effort required to reach the goal from the new node via rapid simulations, rollouts, or neural network-based predictive models. Finally, the update step, known as backpropagation, adjusts the values of all nodes along the traversed path based on the leaf node's evaluation, improving subsequent node selection [25].

The A* algorithm is a heuristic search method that finds an optimal solution when the heuristic function is admissible (and consistent). It operates by iteratively evaluating nodes using the function $f(s) = g(s) + h(s)$, where $g(s)$ is the cost from the start node to state s, and $h(s)$ is an estimate of the cost from s to the goal. The algorithm maintains two sets: *OPEN*, containing nodes to be explored, and *CLOSED*, containing already explored nodes. At each step, A* selects the node from *OPEN* with the lowest f-value. If this node is the goal, the search terminates successfully. Otherwise, the node is expanded, its successors are generated, and their costs are updated if a better path is found before adding them to *OPEN* [25]. This algorithm plays a crucial role, serving as a search engine to identify bio-retrosynthetic planning pathways [25].

The MEEA* algorithm adds a Candidate Set to the *OPEN* and *CLOSED* lists, operating in three vital phases. The simulation phase uses a purely MCTS exploratory policy for look-ahead search without permanent tree expansion. The reached nodes are evaluated, placed in the Candidate Set, and path values are updated. In the selection phase, MEEA* applies A* to the Candidate Set, choosing the state with the lowest $f(s)$ value. Finally, during expansion, the selected node is submitted to the retrosynthetic prediction model, and its children are permanently integrated into the search tree. Tests on the USPTO benchmark revealed that MEEA* achieved strong performance. When replacing Retro* with MEEA*, the success rate for identifying retrosynthetic pathways for NPs compounds increased from 88.66% to 97.68%. Even with a massive advantage, Retro* stagnated at 93.04%, remaining below MEEA*'s standard results [25].

Genome annotation methods Accurate and efficient prediction of enzyme function is essential for genomic annotation workflows. These predictions are essential, and considering the small fraction of enzymes with assigned EC (Enzyme Commission) numbers, numerous tools that use similarity and Machine Learning (ML) techniques have been developed. BLASTp (Basic Local Alignment Search Tool for protein sequences) is the most widely used gold standard tool, but it offers no way of assigning a function to proteins without known homologous sequences. With the advent of Deep Learning (DL), these approaches became better suited for tasks involving large volumes of data. To overcome BLASTp's limitations, DL models began using protein Large Language Models (LLMs) [2].

To evaluate the best approach for genome annotation, Capela et al. (2025) assessed the performance of the ESM2, ESM1b and ProtBERT language models in predicting EC numbers compared to BLASTp, to each other, and to models that rely on one-hot encodings of amino acid sequences [2]. In general, BLASTp outperforms DNN-LLM models on large datasets with homologous sequences available in databases, however, it should be noted that its advantage stems from being the most widely adopted annotation approaches. Despite this, DL models trained with LLMs achieved excellent results, with a very low difference in precision and recall tests compared to BLASTp. In scenarios where the training set had a scarcity of homologous sequences, LLMs outperformed BLASTp. DL models should be used when BLASTp fails to find a sequence with more than 25% identity in the training set, as its performance declines and it is then outperformed by DL models. Combining BLASTp with DNN-LLM models surpasses the performance of either tool used individually, providing an advantage in mainstream annotation routines and more complex tasks [2].

Precursor prediction For precursor prediction, the use of the precursor tool will be essential. The approach to be taken involves an automated ML and DL framework used to optimize molecular analysis tasks. Through its AutoML engine, DeepMol automates the search for the optimal pipeline. It is utilized to solve a chemical and biological problem by automatically identifying the best computational model to predict the precursors of plant secondary metabolites [1].To ensure the identification of the best model, it is described that DeepMol was used to test and compare various approaches, and for this, ML models need to process the data by providing them with a standard representation called molecular fingerprints (FPs) and the main approaches to generate them are Morgan FPs and Layered FPs. Among the approaches designed for NPs, there is a type of FPs that uses substrings known as SMILES, a pre-existing format for writing the two-dimensional structure of a molecule in a simple line of text[1]. Capela et al. (2025) demonstrated that the models identified by DeepMol's AutoML present an advantage over models that use NP-focused FPs. During the testing of the model, precursors were predicted, achieving success for over three-quarters of the compounds. This is an approach that will accelerate the exploration of biosynthetic pathways, potentially expediting the discovery and understanding of natural product biosynthesis [1].

3 Materials and methods

3.1 Selection of phenolic compounds absent from the grapevine GSMM

Initially, the selection of phenolic compounds was performed manually by comparing the supplementary dataset PP2018-RA-00559DR1_Supplemental_Dataset_2.xlsx [6] with the GSMM reference for grapevine, iMS7199. The phenolic secondary metabolites presented in the enhanced dataset that were absent from the iMS7199 GSMM metabolic network were uniquely identified and isolated. For each selected compound, the specific identifier (ID) and molecular chemical structure in SMILES format were collected from the PubChem database. This data was structured and compiled into a CSV (Comma Separated Values) file for subsequent use in developed scripts and computational pipelines. All of this data as well as all the results are available in the repository: `https://github.com/luisbabo/PCsgrapevine-GSMM-DeePlants`.

3.2 Metabolic pathway classification and reaction prediction

The classification of metabolic pathways and the *de novo* prediction of biosynthetic reactions for the previously selected phenolic compounds were performed using the DeePlants tool in Python. The workflow was automated through a structured pipeline in a Snakemake environment, by executing a Snakefile, integrating the pathway_classification and pathway_denovo_classification modules, respectively. The generated predictions classify the metabolic pathways associated with the metabolites using the KEGG (Kyoto Encyclopedia of Genes and Genomes) and PlantCyc reference databases. In the retrosynthesis process (pathway_denovo_classification), reaction predictions are generated, with the Retroformer DL model being used for precursor prediction. The visual representation and reconstruction of the predicted metabolites reactions were generated by executing the pipeline_denovopathways_reactionimages pipeline, paving the way for data analysis and result interpretation.

3.3 Grapevine proteome annotation

The reference grapevine proteome used in this work corresponds to version PN40024.v4, having been obtained in FASTA format from the Integrape database (`https://integrape.eu/wp-content/uploads/2021/11/PN40024.v4_11_05_21.zip`). Through the ProtBERT protein language model, specifically using the predict_with_protbert_from_fasta utility function, it was possible to distinguish protein sequences with enzymatic activity from non-enzymatic sequences, proceeding with the functional annotation of the identified enzymes by predicting and associating their respective EC numbers.

3.4 Use of DeePlants web App to associate enzymes and EC numbers with reactions.

The *de novo* prediction of the biosynthetic pathways of the selected phenolic compounds was carried out with the DeePlants tool, using the Denovo Prediction module. The SMILES of the target compounds were provided as input, and the single-step Retroformer model was used for the retrosynthetic prediction. The Vitis vinifera proteome, as a FASTA file, and the CSV file containing the EC number predictions obtained with ProtBERT were provided in the Protein (FASTA) and EC-solution (CSV) fields, respectively, allowing candidate enzymes to be associated with the predicted reactions.

For the manual verification of individual reactions, the Candidate Enzymes module was used. Each reaction to be analysed was entered in reaction SMILES format (reactants»products), together with the corresponding EC number where known. The grapevine proteome was loaded through the FASTA and CSV files (ProtBERT EC predictions). For each reaction, the tool returned a list of candidate enzymes ranked by an enzyme candidate score, defined as the average of the enzyme–substrate interaction (ESI) probability and the probability of the reaction's EC number classification.

4 Results and discussion

4.1 Metabolic pathways classification and predicted reactions of phenolic compounds

In this work, fourteen phenolic compounds were studied, classified, predicted by retrobiosynthesis and images of these predicted pathways were generated for each of them. The images of the metabolic pathways are provided in the appendix. Among these compounds, benzoic acid and viniferin are the ones discussed in detail, as representative case studies, since they show the range of behaviours observed among those studied. The two compounds correspond to contrasting prediction outcomes, namely a biologically inconsistent route in the case of benzoic acid and on the other hand, a metabolically plausible but enzymatically oversimplified route in the case of viniferin. This analysis thus provides insight into the patterns found for the remaining compounds.

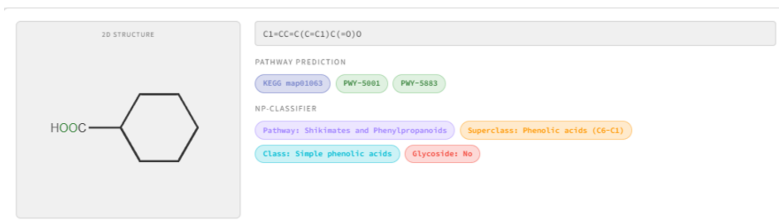

Fig. 1. Pathway associations show the KEGG/PlantCyc/NP-Classifier annotations for the compound benzoic acid.

Through the pathway classifier (Fig. 1), the correct result for its pathway was obtained, namely Shikimates and Phenylpropanoids. Its superclass is likewise correct, being Phenolic acids (C6-C1). As for the class, it is also correct, being Simple phenolic acids, and it is likewise correct in being assigned No for Glycoside. It is in the pathway predictions that errors begin to appear: the three assigned identifiers, KEGG map01063 (biosynthesis of alkaloids derived from the shikimate pathway), PWY-5001 [9] (tetrahydroxyxanthone biosynthesis from benzoate) and PWY-5883 [9] (ephedrine biosynthesis), all correspond to pathways in which benzoate is a precursor or intermediate, rather than the product.

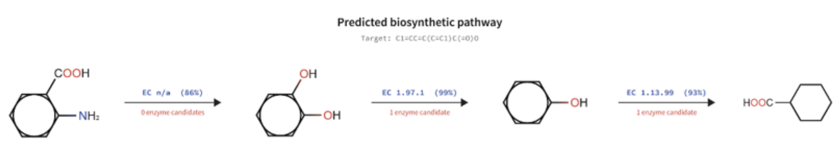

Fig. 2. *De novo* predicted pathways of the metabolite Benzoic acid with assigned EC number and count of candidate enzymes using the DeePlants tool.

Analysing the predicted pathway for benzoic acid, as can be seen in Fig. 2, it proved to be biologically inconsistent with plant metabolism. The precursor proposed by the tool, anthranilic acid, is an intermediate of tryptophan biosynthesis within the chorismate pathway and has no established role in plant phenolic metabolism. Furthermore, the predicted reactions lack biochemical support. The first step has neither an assigned EC number nor candidate enzymes; the second relies on a functionally incompatible enzyme, EC 1.97.1 (chlorate reductase), which is unable to catalyse the removal of hydroxyl groups from aromatic compounds; and the third would require a carboxylation to introduce the carboxyl group, rather than an oxidation as suggested by the assigned classification (EC 1.13.99). It is thus apparent that this pathway reflects a retrosynthetic decomposition driven essentially by the chemical structure of the target, favouring the progressive removal of substituents over transformations with enzymatic precedent in plants.

In contrast, the biosynthesis of benzoic acid in plants is documented in PlantCyc and this pathway starts from L-phenylalanine within the phenylpropanoid pathway. L-phenylalanine is converted into *trans*-cinnamate by phenylalanine ammonia-lyase (PAL, EC 4.3.1.24/4.3.1.25), with the release of ammonium. The subsequent shortening of the side chain by two carbon units can occur through two routes, a CoA-dependent β-oxidative route (benzoate biosynthesis I (CoA-dependent, β-oxidative), PWY-6443 [9]), analogous to fatty-acid β-oxidation and localised in the peroxisome (*trans*-cinnamate $\rightarrow$ cinnamoyl-CoA $\rightarrow$ 3-hydroxy-3-phenylpropanoyl-CoA $\rightarrow$ 3-oxo-3-phenylpropanoyl-CoA $\rightarrow$ benzoyl-CoA) and a non-β-oxidative route (benzoate biosynthesis II (CoA-independent, non-β-oxidative), PWY-6444 [9]), in which *trans*-cinnamate is hydrated and cleaved to benzaldehyde with the release of acetate (EC 4.1.2.-), benzaldehyde being subsequently oxidised to benzoic acid by an NAD^+-dependent benzaldehyde dehydrogenase (EC 1.2.1.28). The pathway predicted by the tool reproduces neither of these routes, further demonstrating a limitation of the retrobiosynthetic approach, since it operates on the target structure without any metabolic knowledge specific to the organism under study. In order to verify a plausible pathway and the number of candidate enzymes for its reactions, the DeePlants Candidate Enzymes tool was used. The pathway benzoate biosynthesis III (CoA-dependent, non-β-oxidative) (PWY-6446 [9]) was replicated, as it is a hybrid variant of the two routes mentioned above. The first reaction, L-phenylalanine $\rightarrow$ *trans*-cinnamate, yielded 11 candidate enzymes (EC 4.3.1). For the following reaction, *trans*-cinnamate $\rightarrow$ cinnamoyl-CoA, 59 candidate enzymes were found (EC 6.2.1). For the next reaction in the pathway, cinnamoyl-CoA $\rightarrow$ (S)-3-hydroxy-3-phenylpropanoyl-CoA, 117 candidate enzymes were found (EC 4.2.1). The subsequent reaction, (S)-3-hydroxy-3-phenylpropanoyl-CoA $\rightarrow$ benzaldehyde, returned a total of 430 candidate enzymes. Finally, the reaction that led to the target, benzaldehyde $\rightarrow$ benzoic acid, yielded 82 candidate enzymes (EC 1.2.1).

Although the pathway classification algorithm for this compound (Fig. 3) did not provide the exact pathway in PlantCyc, but only general flavonoid and phenylpropanoid biosynthesis pathways from the KEGG database, these are nonetheless consistent for the compound in question. The pathway specific

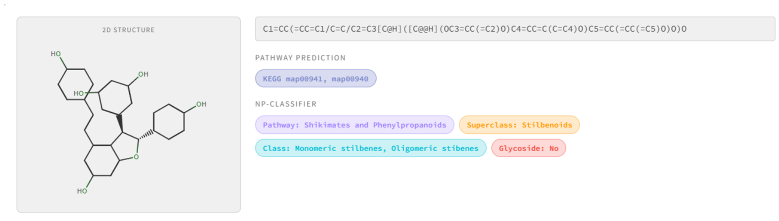

Fig. 3. Pathway associations show the KEGG/PlantCyc/NP-Classifier annotations for the compound Viniferin

to stilbenoids (map00945), in which resveratrol and its oligomers are included, was not assigned, however. Through a literature search, it was possible to identify the resveratrol pathway in the PlantCyc database, with the ID PWY-84 [9] .

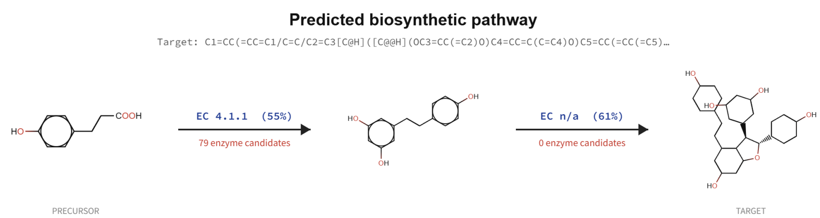

Fig. 4. *De novo* predicted pathways of the metabolite Viniferin with assigned EC number and count of candidate enzymes using the DeePlants tool

Analysing Fig. 4, which concerns the predicted pathway for the PC viniferin, and in contrast to the case of benzoic acid, this pathway starts from the correct precursor and correctly reproduces the biosynthetic logic at the metabolite level, namely *p*-coumaric acid $\rightarrow$ resveratrol $\rightarrow$ viniferin, reflecting the real phenylpropanoid $\rightarrow$ stilbene $\rightarrow$ oligostilbene route. In this case, the limitations lie at the enzymatic and mechanistic level. The first step was reduced to a decarboxylation (EC 4.1.1), whereas the formation of resveratrol requires the activation of *p*-coumarate to *p*-coumaroyl-CoA (EC 6.2.1.12) and the subsequent condensation with three units of malonyl-CoA by stilbene synthase (EC 2.3.1.95), a process that builds the resorcinol ring *de novo*. The second step, the oxidative dimerisation of resveratrol, was correctly predicted in terms of the transformation, but no enzyme was assigned to it (EC n/a). It is known that resveratrol is converted into viniferin (δ-viniferin) through peroxidase-mediated oxidative coupling in the presence of H_2O_2 [15], catalysed by class III peroxidases (EC 1.11.1.7). In order to analyse the assignment of EC numbers and candidate enzymes, the Candidate Enzymes tool was used. For the reaction *p*-coumarate → *p*-coumaroyl-CoA, a total of 59 candidate enzymes was obtained (EC 6.2.1). For the following reaction, *p*-coumaroyl-CoA + malonyl-CoA → resveratrol, 430 candidate enzymes were predicted (EC 2.3.1). Finally, for the reaction resveratrol → δ-viniferin, 125 candidate enzymes were predicted (EC 1.11.1).

These results suggest that, for polyketide-derived skeletons, the tool tends to capture the relationship between metabolites but to simplify or omit multi-step enzymatic transformations.

5 Conclusion

In this work, the DeePlants framework was evaluated for the classification and retrobiosynthetic prediction of phenolic secondary metabolites in grapevine (*Vitis vinifera*). Fourteen phenolic compounds absent from the iMS7199 GSMM were selected from a dataset, classified with respect to their pathways, and subjected to retrobiosynthetic reaction prediction using the Retroformer model, with candidate enzymes predicted from the grapevine proteome (PN40024.v4) annotated with ProtBERT.

The detailed analysis of benzoic acid and viniferin showed that the structural classification was generally accurate, correctly placing both compounds within the shikimate and phenylpropanoid context, whereas the pathway and reaction predictions revealed consistent limitations. Compounds tended to be associated with adjacent or mechanistically analogous pathways, or with pathways in which they appear only as a precursor or intermediate, rather than the pathway dedicated to their biosynthesis. The predicted routes ranged from biologically inconsistent, as for benzoic acid, whose proposed precursor and assigned enzymes have no precedent in plant phenolic metabolism, to metabolically plausible but enzymatically oversimplified, as for viniferin, where the correct *p*-coumaric acid → resveratrol → viniferin logic was reproduced but multi-step transformations were collapsed or left unassigned. In all cases, manual curation against curated pathways and the grapevine proteome was required to obtain a sound route.

Overall, DeePlants proved valuable for rapidly generating structural classifications and hypotheses on biosynthetic routes and candidate enzymes, but its predictions still require curation and organism-specific biochemical knowledge before reliable use, while manually validated pathways provide a curated basis that could, in future work, support the expansion of phenolic secondary metabolism underrepresented in grapevine genome-scale metabolic models.

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A *De novo* predicted pathways of the studied phenolic compounds

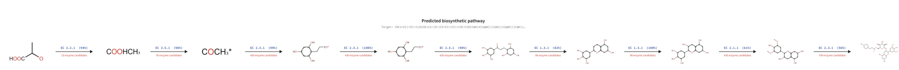

Fig. 5. *De novo* predicted pathway of the metabolite malvidin 3-(6”-p-coumarylglucoside) with assigned EC number and count of candidate enzymes using the DeePlants tool.

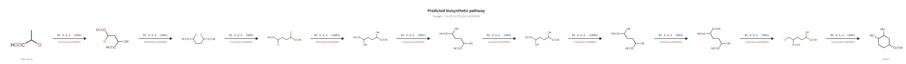

Fig. 6. *De novo* predicted pathway of the metabolite 3,4-dihydroxybenzoic acid with assigned EC number and count of candidate enzymes using the DeePlants tool.

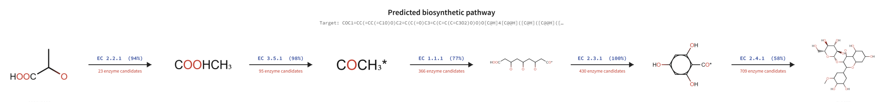

Fig. 7. *De novo* predicted pathway of the metabolite laricitrin-3-glucoside with assigned EC number and count of candidate enzymes using the DeePlants tool.

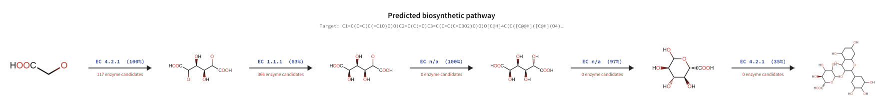

Fig. 8. *De novo* predicted pathway of the metabolite myricetin-3-O-glucuronide with assigned EC number and count of candidate enzymes using the DeePlants tool.

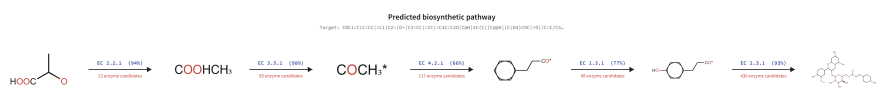

Fig. 9. *De novo* predicted pathway of the metabolite peonidin 3-(6”-p-coumaroylglucoside) with assigned EC number and count of candidate enzymes using the DeePlants tool.

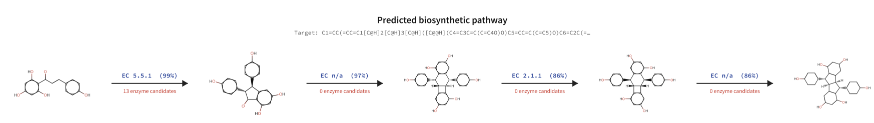

Fig. 10. *De novo* predicted pathway of the metabolite pallidol with assigned EC number and count of candidate enzymes using the DeePlants tool.

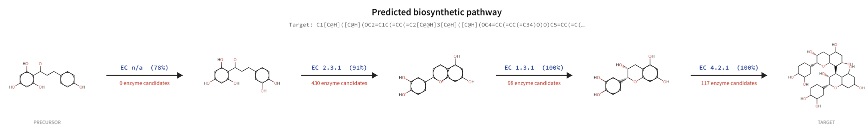

Fig. 11. *De novo* predicted pathway of the metabolite procyanidin B2 with assigned EC number and count of candidate enzymes using the DeePlants tool.

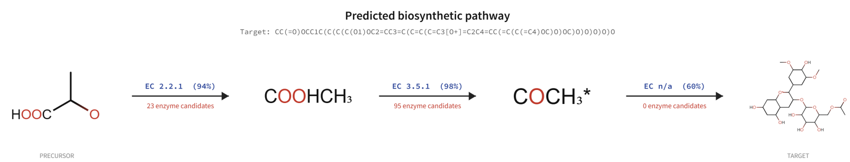

Fig. 12. *De novo* predicted pathway of the metabolite malvidin 3-(6”-acetylglucoside) with assigned EC number and count of candidate enzymes using the DeePlants tool.

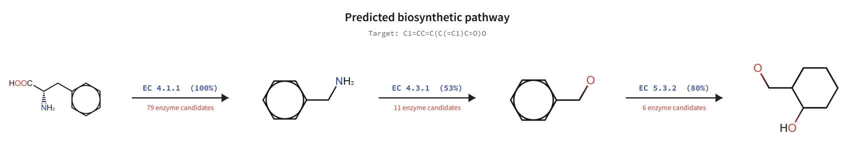

Fig. 13. *De novo* predicted pathway of the metabolite salicylaldehyde with assigned EC number and count of candidate enzymes using the DeePlants tool.

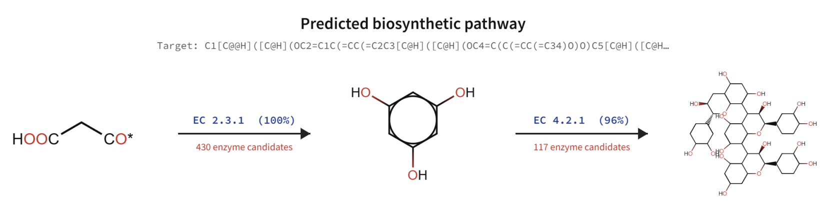

Fig. 14. *De novo* predicted pathway of the metabolite procyanidin trimer with assigned EC number and count of candidate enzymes using the DeePlants tool.

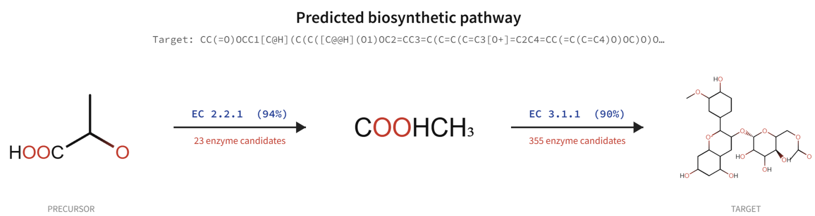

Fig. 15. *De novo* predicted pathway of the metabolite peonidin 3-(6”-acetylglucoside) with assigned EC number and count of candidate enzymes using the DeePlants tool.

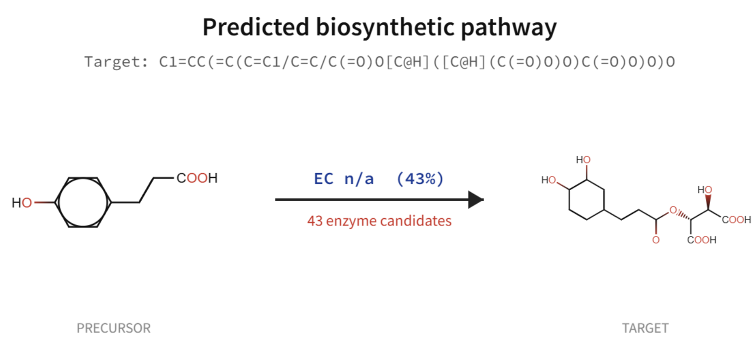

Fig. 16. *De novo* predicted pathway of the metabolite caftaric acid with assigned EC number and count of candidate enzymes using the DeePlants tool.